Disaster Damage Detection via Satellite Imagery

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***Abstract*—Post-disaster building damage maps are a first input to rescue logistics, but automated assessment from satellite imagery remains difficult: class imbalance is severe, intermediate damage categories are visually ambiguous, and models trained on known disasters generalize poorly to new ones. We take a two-phase approach: Phase 1 fine-tunes a foundation model encoder on pre-disaster imagery for building localization, and Phase 2 extends the same encoder into a dual-branch architecture that processes pre- and post-disaster image pairs for five-class damage grading. We first reproduced two competitive baselines to calibrate our implementation: the xView2 first-place ensemble [1], reproduced at F1 = 0.788 (reported: 0.811), and DisasterAdaptiveNet [2], reproduced at F1 = 0.533 on the event-based split (reported: 0.541). For Phase 1, we evaluated several geospatial foundation models and found that DeepLabV3+ [3] with an ImageNet-pretrained ResNet-50 backbone achieves the best localization: Dice = 0.877, IoU = 0.781. The Phase 2 system incorporates asymmetric change-signal fusion, Earth Mover's Distance ordinal loss, and a dual-level imbalance correction combining weighted tile sampling with inverse-frequency focal loss. Our best model achieves a damage mean F1 of 0.7873 and localization F1 of 0.8752, yielding an overall score of 0.8136 ($0.3\times$Loc F1 $+ 0.7\times$Dmg F1), which exceeds the xView2 winner's score of 0.8112 under the same metric. The code is available at https://github.com/MulyaP/Disaster-Damage-Detection/tree/submission.**

I. Introduction

A. Motivation and Problem Statement

Massive natural disasters cause large-scale infrastructure damage every year, affecting millions of lives and incurring enormous economic losses. Rapid building damage assessment is the first step in disaster response: it determines movement patterns and enables targeted allocation of resources such as temporary shelter, medical aid, and reconstruction. Traditional field-based inspection is slow, dangerous, and impossible to scale to large or inaccessible areas.

The combination of remote sensing and deep learning has made automated damage assessment practical. Datasets such as xBD [4] provide co-registered pre- and post-disaster satellite imagery annotated with per-building damage severity, enabling data-driven models. The dominant methodology frames assessment as a two-stage process: first localize building footprints, then classify each building's damage level into four ordinal categories (no damage, minor, major, destroyed).

Several fundamental challenges persist across the field. First, the class distribution is heavily skewed: the majority of buildings are undamaged, and minor and major damage together account for only a small fraction of labeled pixels. Second, the intermediate damage categories are visually ambiguous — a minor-damage building and a no-damage building may look nearly identical from 0.5 m altitude — making fine-grained classification unreliable. Third, models trained on one set of disaster events generalize poorly when tested on unseen disaster types. Our work targets all three of these challenges directly.

B. Related Work

This section surveys the datasets, current state-of-the-art methods, and foundation models most relevant to our work.

1) Datasets and Benchmarks: Large annotated datasets have driven progress in building damage detection. Before 2019, most methods trained and tested on small datasets tied to single disaster events, making cross-disaster generalization hard to evaluate [2].

Gupta et al. [4] released xBD alongside the xView2 challenge. xBD contains 850,736 labeled building polygons across 19 disaster types at 0.5 m/pixel resolution, with co-registered pre- and post-disaster RGB pairs at 1024×1024 pixels covering 45,362 km^2. Each polygon carries a four-level damage label from the Joint Damage Scale: no damage, minor damage, major damage, and destroyed. The xView2 challenge defined a harmonic mean F1 score combining localization and classification that the field has used as its standard metric since. Despite newer additions such as BRIGHT [5], which pairs optical imagery with SAR for cloud-robust assessment, xBD remains the largest multi-hazard benchmark and the primary reference for comparing methods.

2) Deep Learning Approaches: Early deep learning work was hazard-specific and did not generalize across disaster types [2], [6]. Two architectural patterns emerged: the U-Net encoder-decoder [7] with skip connections became the standard for building segmentation, and a two-stage pipeline — localize buildings first, then classify damage using bitemporal features — became the dominant design. The xView2 winning solution [1] applied this pipeline with a large ensemble of ResNet and SENet variants, reaching a challenge score of 0.812.

Subsequent work moved toward joint architectures that optimize both tasks through shared representations. Zheng et al. [8] introduced ChangeOS, a dual-stream network with a shared encoder and object-based post-processing that generalized to the 2020 Beirut explosion — a disaster unseen during training. Siamese and dual-stream architectures that compare pre- and post-disaster features have since consistently outperformed single-image baselines on damage classification.

Vision Transformers [9] improved on CNNs by capturing long-range spatial dependencies relevant to understanding large damaged areas as coherent structures. Chen et al. [10] introduced ChangeMamba, a state space model that matches transformer performance at lower computational cost — an important property for large-image inference.

A persistent problem is the gap between challenge-split and event-based performance. Hafner et al. [2] showed that holding out entire disaster events from training causes consistent drops, and that minor and major damage categories have much lower F1 scores than no-damage and destroyed categories across all methods.

3) Foundation Models for Remote Sensing: Large pre-trained models have changed transfer learning in remote sensing. He et al. [11] showed that masked autoencoding is an effective self-supervised objective for vision transformers. Cong et al. [12] adapted this to satellite imagery with Sat-MAE, pretraining on the multi-spectral, multi-temporal fMoW dataset. Wang et al. [13] proposed RVSA, adapting plain vision transformers for remote sensing with rotated varied-size window attention pretrained on MillionAID. Bastani et al. [14] released Satlas with a built-in segmentation head capable of pixel-level predictions without a separate decoder. Guo et al. [15] trained SkySense on 21.5 million temporal optical and SAR sequences, reaching state-of-the-art results across a wide set of remote sensing benchmarks.

In the general vision domain, Oquab et al. [16] trained DINOv2 on 142 million curated images using self-distillation, producing features that transfer well to dense prediction tasks. Xie et al. [17] proposed SegFormer, a transformer segmentation model with a lightweight all-MLP decoder that serves as a reliable fine-tuning baseline. Chen et al. [3] introduced DeepLabV3+, which combines atrous spatial pyramid pooling (ASPP) with an encoder-decoder structure for multi-scale dense prediction, and Wang et al. [18] proposed HRNet, which keeps high-resolution feature representations active throughout the network.

Kirillov et al. [19] introduced SAM, training a large vision transformer on over one billion masks to produce a general-purpose promptable segmentation model useful for post-disaster scenarios where labeled data for a new event may not yet exist. Gong et al. [20] built CrossEarth around domain generalization, pretraining on diverse sensors and imaging conditions to improve cross-domain transfer.

4) Open Problems: Two problems remain open. First, all competitive methods need large volumes of labeled data from known disaster types; cross-disaster generalization is still poor [2]. Second, minor and major damage classification lags behind localization and destroyed-building detection across all methods, suggesting that appearance-based optical features alone are insufficient for fine-grained damage grading. Our work directly addresses the second problem through ordinal loss design and imbalance correction, and partially addresses the first by leveraging foundation model pretraining to reduce reliance on task-specific labeled data.

II. Reproducibility Studies

Before building our own system we reproduced two competitive baselines to establish calibrated reference points and understand where each method's performance comes from.

A. xView2 First-Place Solution [1]

The xView2 winning solution trains a two-stage pipeline — building localization followed by per-building damage classification — using an ensemble of 42 models across four backbone architectures: ResNet-34, ResNet-50, DPN-92, and SENet-154. Each backbone is trained with different random seeds and data augmentation configurations; predictions are ensemble-averaged at inference. The reported challenge score is 0.811, with a damage-only F1 of 0.779.

a) Our reproduction.: We reproduced the solution using the released GitHub code [1], making only the following changes: (1) we trained 12 models instead of 42 due to time and GPU resource constraints; (2) all training ran on a single NVIDIA L40S GPU instead of two GPUs; (3) the Apex mixed-precision library was ported to native PyTorch `torch.amp`, and deprecated NumPy operations were updated. No changes were made to model architecture, dataset splits, hyperparameters, or loss functions.

Our result was $\mathrm{F1} = \mathbf{0.788}$ (reported: 0.811). The gap of 0.023 is consistent with ensemble size: training $3\times$ fewer models reduces ensemble diversity and therefore the performance ceiling. This is not a fundamental reproducibility failure but an expected resource-constrained approximation.

B. DisasterAdaptiveNet [2]

DisasterAdaptiveNet (DAN) is a Siamese encoder-decoder that conditions its feature extraction on a disaster-type embedding via Feature-wise Linear Modulation (FiLM), allowing the model to adapt its representations to the specific hazard at inference time. The authors report two sets of scores: 0.770 on the original xBD random split (where training and test images can share the same disaster event) and 0.541 on their proposed event-based split (where entire disaster events are held out from training).

a) Our reproduction.: We used the authors' released pretrained weights directly, processed the xBD masks from JSON polygon annotations, and evaluated using the xView2 scoring script. We tested both the DAN-event model and the authors' strong baseline.

Under the event-based split our DAN-event model scored **0.533** (reported: 0.541) and the strong baseline scored **0.478** (reported: 0.483). Both gaps are small and are attributable to minor differences in data preprocessing and the evaluation environment. An important finding is that both models struggle with the minor and major damage classes ($\mathrm{F1} \approx 0.25$–0.34), consistent with the broader literature.

b) Split methodology insight.: The event-based split is the realistic evaluation: it tests whether a model trained on known disasters generalizes to a new, unseen event — exactly the deployment scenario. The random split is optimistic because training and test images can come from the same tornado or hurricane; the model has already seen buildings from that location under similar conditions. Phase 2 training uses the event-based xBD hold split for consistency.

III. Methodology

Our approach mirrors the two-stage design found in the literature, but differs in three ways: (1) we select the localization backbone from a comparison of foundation models rather than assuming ImageNet pretraining is sufficient; (2) we use an asymmetric dual-branch fusion that provides an explicit learned change signal for damage prediction; and (3) we combine ordinal loss design with a two-level data imbalance correction.

A. Phase 1 — Building Localization

The goal of Phase 1 is to produce a per-pixel building mask $\hat{M} \in [0,1]^{H\times W}$ from a single pre-disaster image $\mathbf{I}_{pre} \in \mathbb{R}^{H\times W\times 3}$, using a foundation encoder $\mathcal{F}_\theta$ and segmentation decoder $\mathcal{D}_\phi$:

$$\hat{M} = \mathcal{D}_\phi(\mathcal{F}_\theta(\mathbf{I}_{pre}))\,. \tag{1}$$

Because building pixels account for only ∼6% of xBD, we train with a combination of binary cross-entropy and Dice loss to handle the foreground–background imbalance:

$$\mathcal{L}^{\text{P1}}_{\text{loc}} = \mathcal{L}_{\text{BCE}}(\hat{M}, M) + \big(1 - \tfrac{2\,|\hat{M}\cap M|}{|\hat{M}|+|M|}\big). \tag{2}$$

The selected Phase 1 encoder weights θ^* are then transferred to Phase 2.

B. Phase 2 — Dual-Branch Damage Classification

Phase 2 takes a matched image pair and predicts an ordinal five-class damage label for each pixel. The encoder from Phase 1 is shared across both branches; both images are processed by the same ResNet-50 backbone at two spatial scales.

a) Encoder.: Both $\mathbf{I}_{pre}$ and $\mathbf{I}_{post}$ are passed through the shared encoder $\mathcal{F}_\theta$, which extracts features at two scales:

$$(\mathbf{L}_{pre}, \mathbf{H}_{pre}) = \mathcal{F}_\theta(\mathbf{I}_{pre}), \quad (\mathbf{L}_{post}, \mathbf{H}_{post}) = \mathcal{F}_\theta(\mathbf{I}_{post}), \tag{3}$$

where $\mathbf{L} \in \mathbb{R}^{B\times 256\times H/4\times W/4}$ is the low-level feature map (after ResNet `layer1`, stride 4) and $\mathbf{H} \in \mathbb{R}^{B\times 1024\times H/16\times W/16}$ is the high-level feature map (after dilated `layer3`, effective stride 16).

b) Asymmetric fusion.: Pre- and post-disaster features are fused differently for the two decoder heads. *Localization* uses only appearance:

$$\mathbf{L}_{loc} = \text{cat}(\mathbf{L}_{pre}, \mathbf{L}_{post}), \quad \mathbf{H}_{loc} = \text{cat}(\mathbf{H}_{pre}, \mathbf{H}_{post}). \tag{4}$$

Damage classification additionally incorporates an explicit change signal and applies learned channel compression to avoid inflating the decoder width:

$$\mathbf{H}_{dmg} = W_H[\text{cat}(\mathbf{H}_{pre},\, \mathbf{H}_{post},\, \mathbf{H}_{post} - \mathbf{H}_{pre})]\,, \tag{5}$$

$$\mathbf{L}_{dmg} = W_L[\text{cat}(\mathbf{L}_{pre},\, \mathbf{L}_{post},\, \mathbf{L}_{post} - \mathbf{L}_{pre})]\,, \tag{6}$$

where W_H and W_L are 1×1 convolutions that compress the 3×-width concatenation back to 2×-width, preserving decoder capacity while making the change signal explicitly available. The rationale is that localization depends on appearance (buildings look the same before and after), while damage requires change (a destroyed building looks different from its pre-disaster state).

c) Decoders.: Both heads use the same DeepLabV3+-style decoder: ASPP on the high-level features, a 48-channel projection of the low-level features, concatenation and two 3×3 convolutions, followed by a 1×1 classification convolution. The localization head outputs one logit per pixel; the damage head outputs five logits.

C. Training and Optimization

a) Phase 1 localization loss.: Phase 1 minimises a sum of binary cross-entropy and Dice loss (see Section III-A).

b) Phase 2 localization loss.: The joint model's localization head is trained with Binary Focal Loss [21] ($\gamma = 2$), which down-weights easy background pixels automatically:

$$\mathcal{L}_{\text{loc}} = -\frac{1}{N}\sum_i \Big[y_i\,(1-\hat{p}_i)^\gamma \log \hat{p}_i \;+\; (1-y_i)\,\hat{p}_i^\gamma \log(1-\hat{p}_i)\Big]. \tag{7}$$

c) Damage classification loss.: Multi-class Focal Loss [21] with per-class inverse-frequency weights w and background ignored (`ignore_index`$=0$):

$$\mathcal{L}_{\text{dmg}} = -\frac{1}{N}\sum_i w_{y_i}(1-\hat{p}_{y_i})^\gamma \log \hat{p}_{y_i}. \tag{8}$$

Class weights are computed as $w_c = \frac{n_{\text{total}}}{n_c}$, normalised so $\sum_c w_c = 4$.

d) EMD ordinal loss.: Standard cross-entropy treats all misclassifications equally, but damage severity is ordinal: predicting "minor" when the truth is "major" is a smaller error than predicting "destroyed". We add an Earth Mover's Distance loss that penalises predictions in proportion to their ordinal distance from the ground truth:

$$\mathcal{L}_{\text{EMD}} = \frac{1}{N}\sum_i \sum_{c=1}^{C} \big(\text{CDF}(\hat{p}_i)_c - \text{CDF}(\mathbf{e}_{y_i})_c\big)^2, \tag{9}$$

where $\text{CDF}(\cdot)_c$ is the cumulative distribution function up to class c, $\hat{p}_i$ is the predicted probability vector, and $\mathbf{e}_{y_i}$ is the one-hot target.

e) Total loss.:

$$\mathcal{L} = \lambda_{\text{loc}}\,\mathcal{L}_{\text{loc}} + \lambda_{\text{dmg}}\,\mathcal{L}_{\text{dmg}} + \lambda_{\text{EMD}}\,\mathcal{L}_{\text{EMD}}, \tag{10}$$

with $\lambda_{\text{loc}} = 1.0$, $\lambda_{\text{dmg}} = 1.0$, $\lambda_{\text{EMD}} = 0.5$.

f) Class imbalance.: Building damage datasets exhibit imbalance at two levels: most image tiles contain no or little damage, and within building pixels the "no damage" class dominates. We address both levels simultaneously.

Tile-level: A `WeightedRandomSampler` assigns each 512×512 tile a sampling weight based on its damage content: tiles containing any minority damage class (minor or major)

TABLE I
REPRODUCIBILITY RESULTS. VALUES IN PARENTHESES ARE REPORTED IN THE ORIGINAL PAPERS.

Method	Split	Ours	Reported
xView2 First Place [1]	xBD random	0.788	0.811
DisasterAdaptiveNet [2]	xBD event-based	0.533	0.541
Strong Baseline [2]	xBD event-based	0.478	0.483

receive weight 3×, and background-dominated or no-damage-only tiles receive weight 1×. This ensures every training batch contains examples of rare damage classes.

Pixel-level: Inverse-frequency class weights in the Focal Loss additionally up-weight rare damage categories at the loss computation level.

g) Training schedule.: **Phase 1** trains for 30 epochs with AdamW [22] (lr = 1e-4) and CosineAnnealingLR. The best checkpoint (by mean of validation IoU and Dice, $(\mathrm{IoU} + \mathrm{Dice})/2$) is saved and its encoder weights are transferred to Phase 2.

Phase 2 uses differential learning rates: the shared encoder is updated at $0.5\times$ the decoder learning rate to preserve the localization features learned in Phase 1. Training runs for 50 epochs with a 5-epoch linear warmup followed by CosineAnnealingLR over the remaining 45 epochs. Mixed-precision training (`torch.amp`) is used throughout. All training was done in linux environment using the NVIDIA L40S GPU.

Data augmentation is applied consistently to the image pair: random horizontal/vertical flips, random 90° rotations, and brightness/contrast jitter. Geometric transforms are applied identically to both pre- and post-disaster tiles to preserve spatial alignment.

IV. PHASE 1 — FOUNDATION MODEL COMPARISON

A. Models Surveyed

We surveyed a selection of foundation model families as candidates for the Phase 1 encoder, focusing on models whose pretraining data is compatible in resolution with xBD (0.5 m/pixel) and whose weights are publicly available. Table II summarizes the surveyed models.

B. Fine-Tuning Setup

All experiments target binary building segmentation from pre-disaster images. xBD images are 1024×1024 pixels; models with smaller native input sizes (maximum 512) are fed 512×512 tiles with a minimum building-pixel ratio of 1% (tiles with fewer building pixels are discarded), and predictions are stitched at inference. Building pixels account for only $\sim$6% of the total pixel area, so Focal Loss or BCE is used throughout.

Models with randomly initialized task heads (SegFormer, Satlas) followed a two-stage schedule: the head alone was trained for 5 epochs first, then all layers were trained end-to-end with differential learning rates (encoder: 1e-5, head/upsample: 1e-4).

C. Phase 1 Results

a) DeepLabV3+: achieved the best performance at Dice = 0.877, IoU = 0.781 with an ImageNet-pretrained ResNet-50 backbone. The fMoW-pretrained variant (Dice = 0.873) slightly underperformed, which was unexpected: fMoW is an overhead satellite imagery dataset and should offer better domain match. We attribute this to fMoW weights being optimized for scene-level classification rather than dense pixel prediction, making the feature geometry less suited to segmentation.

b) DINOv2: achieved Dice = 0.874 with tiled 512×512 crops, placing it second — remarkable given that it was pre-trained exclusively on natural images. An initial run with naive downsampling (1024→512) produced Dice = 0.536; switching to tiling recovered the full performance, confirming that resolution-preserving preprocessing is critical for this dataset.

c) Satlas: reached Dice = 0.819. Visual inspection revealed over-localization: the model predicts building pixels in regions with no buildings, and merges nearby structures whose boundaries should remain distinct. This is consistent with Satlas pretraining on coarser (10 m/px Sentinel-2) data, where fine boundary discrimination is less important.

d) SAM: was evaluated in prompted mode using ground-truth building polygon centroids as point prompts, which provides an optimistic upper bound (Dice = 0.869) but is not practical during deployment when ground truth is unavailable.

e) SegFormer: reached only Loc F1 = 0.628. As the only model pretrained solely on non-geospatial data (ADE20K) without a dedicated remote sensing segmentation objective, this result is expected.

f) Selection.: DeepLabV3+ with ImageNet backbone was selected for Phase 2 based on its top Dice score, computational efficiency, and the well-understood ASPP multi-scale architecture. Its weights after Phase 1 fine-tuning initialize the Phase 2 encoder.

V. PHASE 2 — DAMAGE CLASSIFICATION

We trained the JointDamageNet architecture (Section III-B) on xBD `tier1+tier3`, validating on the `hold` split. The overall score is computed as a weighted sum $0.3 \times \mathrm{Loc\,F1} + 0.7 \times \mathrm{Dmg\,F1}$, where Dmg F1 is the mean F1 across the four damage classes (no-damage, minor, major, destroyed). The best validation checkpoint was reached at epoch 23:

- Overall score: **0.8136**
- Damage mean F1: **0.7873**
- Localization F1: **0.8752**
- Minor-damage F1: **0.5891**

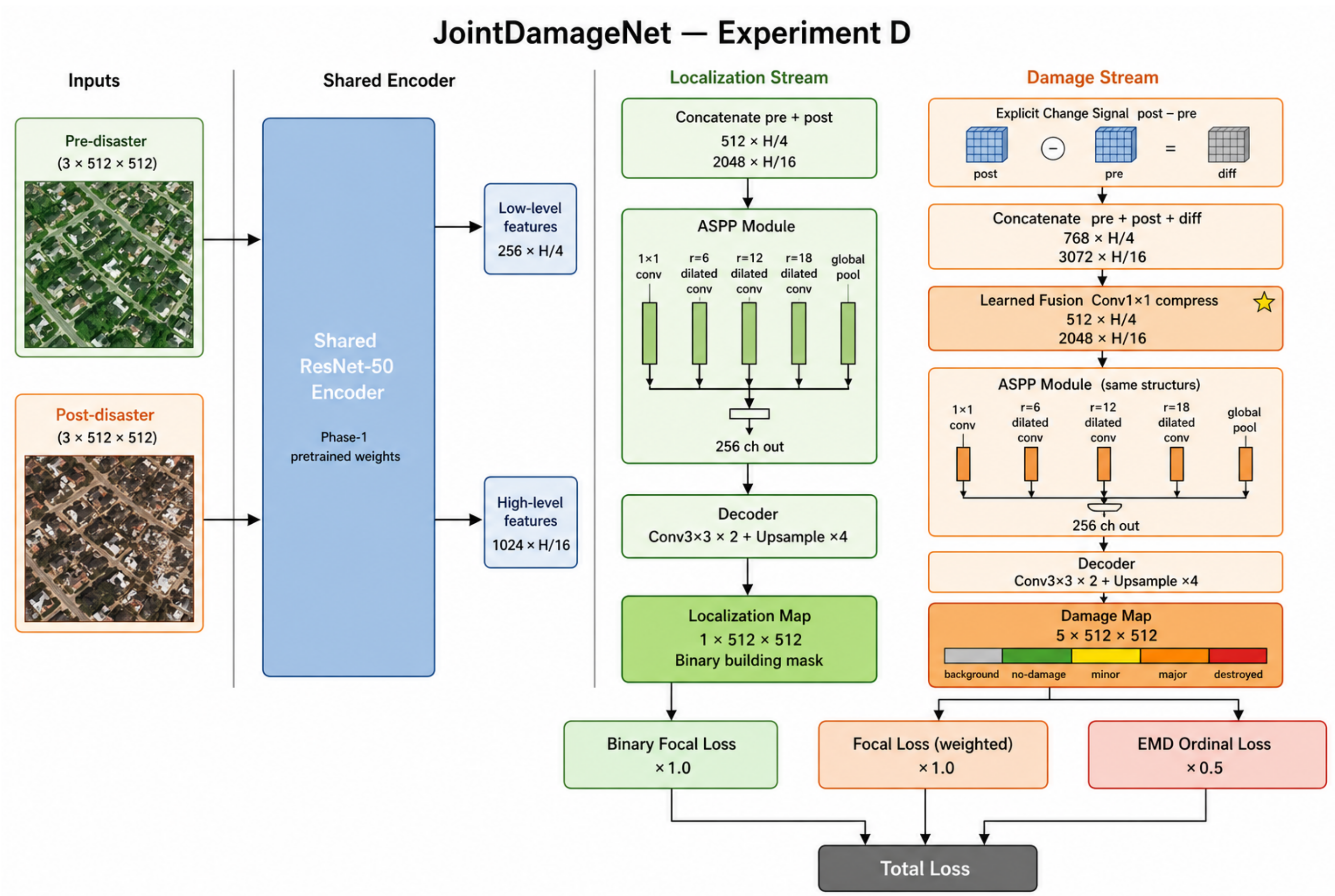

Fig. 1. Two-phase pipeline. Phase 1 fine-tunes DeepLabV3+ on pre-disaster images for building localization. Phase 2 extends the encoder into a dual-branch architecture: pre- and post-disaster images share the encoder, features are fused asymmetrically (simple concatenation for localization; concatenation + explicit difference + learned compression for damage), and two decoder heads produce localization and damage maps jointly. Image generated by ChatGPT.

- Major-damage F1: **0.6124**
- No-damage / Destroyed: $>0.89 / >0.92$

VI. Results and Discussion

Our overall score of 0.8136 places us competitively against ChangeMamba (0.8141) and above the xView2 winner (0.8112) and all other compared methods. Our localization F1 of 0.8752 is the highest across all methods in the table. On the hardest class, minor-damage F1 of 0.589 substantially improves over DisasterAdaptiveNet [2]'s 0.253 on the same evaluation split.

The results also reveal several design insights worth noting.

a) Asymmetric fusion.: Providing the explicit change signal ($\mathbf{H}_{post} - \mathbf{H}_{pre}$) with learned compression to the damage decoder, while leaving the localization head to use simple concatenation, allows each head to receive exactly the information it needs: appearance for localization, change for damage.

b) Ordinal loss.: Standard cross-entropy treats predicting "no damage" for a "destroyed" building the same as predicting "major damage" for it. The EMD loss penalizes predictions in proportion to their ordinal distance from the ground truth, which directly addresses the severity-of-error structure of the damage classification task.

c) Two-level imbalance correction.: Focal loss alone does not solve the problem when rare damage classes appear in too few training batches regardless of per-pixel weighting. Combining tile-level weighted sampling with pixel-level inverse-frequency weights ensures every batch contains rare-class examples and that those examples receive proportionally higher gradient signal.

d) Foundation model choice for Phase 1.: DeepLabV3+ with an ImageNet-pretrained backbone outperforms geospatial-native models on the localization task. DINOv2 comes close (Dice 0.874 vs. 0.877) despite being pretrained on natural images. Models pretrained for scene classification (fMoW) do not automatically transfer to dense pixel-level segmentation, even when the pretraining domain matches.

VII. Conclusion

We built a two-phase system for building damage detection from satellite imagery that addresses the challenges of class imbalance, ordinal severity prediction, and change-signal extraction.

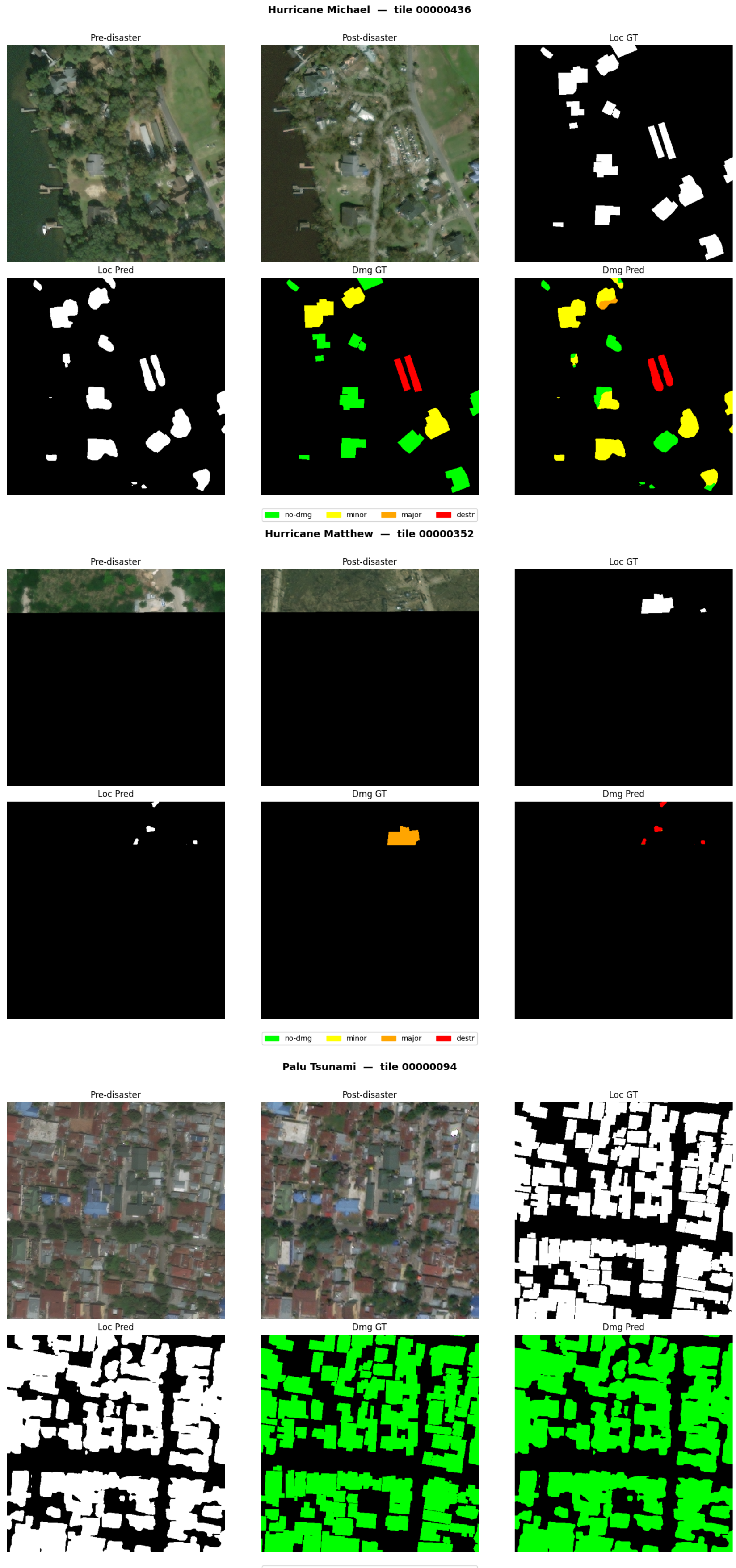

Fig. 2. Sample predictions by our model. **Top (Hurricane Michael):** the model confuses minor and major damage categories. **Middle (Hurricane Matthew):** poor building localization propagates to poor damage classification. **Bottom (Palu Tsunami):** a representative well-localized prediction.

TABLE II
FOUNDATION MODELS SURVEYED FOR THE PHASE 1 LOCALIZATION TASK.

Model	Pretrained On	Native Resolution	Weights
DINOv2 [16]	LVD-142M (natural images)	general	✓
SatMAE [12]	fMoW RGB	0.3–1 m/px	✓
RVSA [13]	MillionAID	0.3–30 m/px	✓
Satlas [14]	Satlas-pretrain (NAIP + Sentinel-2)	0.5–10 m/px	✓
SkySense [15]	HSROIs + Sentinel-1/2 (21.5M sequences)	0.5–30 m/px	✓
SegFormer [17]	ADE20K	general	✓
DeepLabV3+ [3]	ImageNet / fMoW	general	✓
SAM [19]	SA-1B	general	✓
CrossEarth [20]	Multi-domain remote sensing	0.5–11.4 m/px	✓

TABLE III
LOCALIZATION RESULTS ON THE XBD HOLD VALIDATION SPLIT. HIGHER IS BETTER.

Model	Variant / Backbone	Val. Dice	Val. IoU
DeepLabV3+ [3]	ImageNet pretrain	**0.877**	**0.781**
DINOv2 [16]	LVD-142M	0.874	0.776
DeepLabV3+ [3]	fMoW pretrain	0.873	0.775
SAM* [19]	ViT-H (GT prompts)	0.869	0.768
Satlas [14]	Aerial SwinB-SI	0.819	0.747
SegFormer [17]	MiT-B2 (ADE20K)	0.628	—

*SAM uses ground-truth polygon centroids as prompts (upper bound; not practical at deployment).

TABLE IV
COMPARISON AGAINST COMPETITIVE BASELINES ON XBD. SCORES FOR PRIOR METHODS FROM HAFNER ET AL. [2]'S UNIFIED EVALUATION; OUR SCORES ARE ON THE XBD HOLD SPLIT.

Method	Overall F1	Damage F1	Loc F1
ChangeMamba [10]	0.8141	0.7884	0.8738
xView2 1st Place [1]	0.8112	0.7887	0.8635
ChangeOS [8]	0.7857	0.7564	0.8541
DamFormer†	0.7702	0.7281	0.7281
Ours	**0.8136**	**0.7873**	**0.8752**

†No separate publication; scores from Hafner et al. [2]'s benchmark.

Phase 1 compared several geospatial foundation models for building localization, finding that DeepLabV3+ with ImageNet pretraining achieves Dice = 0.877 and outperforms geospatial-specific alternatives. Phase 2 used a dual-branch architecture with asymmetric change-signal fusion, EMD ordinal loss, and dual-level imbalance correction, achieving a damage mean F1 of 0.7873 and an overall score of 0.8136, with the highest localization F1 (0.8752) among compared methods and a minor-damage F1 of 0.589 — substantially higher than the 0.253 reported by DisasterAdaptiveNet on the same class.

a) Limitations and future work.: Worthwhile next steps include combining the Phase 2 architecture with a geospatial-native encoder (DINOv2 or Satlas) once the resolution preprocessing issues are resolved, testing on the event-based DisasterAdaptiveNet split to measure cross-disaster generalization, and integrating SAR imagery from datasets like BRIGHT for cloud-robust all-weather assessment.

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